

where talent goes next:

MAPPING ECONOMIC OPPORTUNITY
THROUGH SOCIAL CAPITAL AND MOBILITY

PRESENTED BY:
MERCY EME



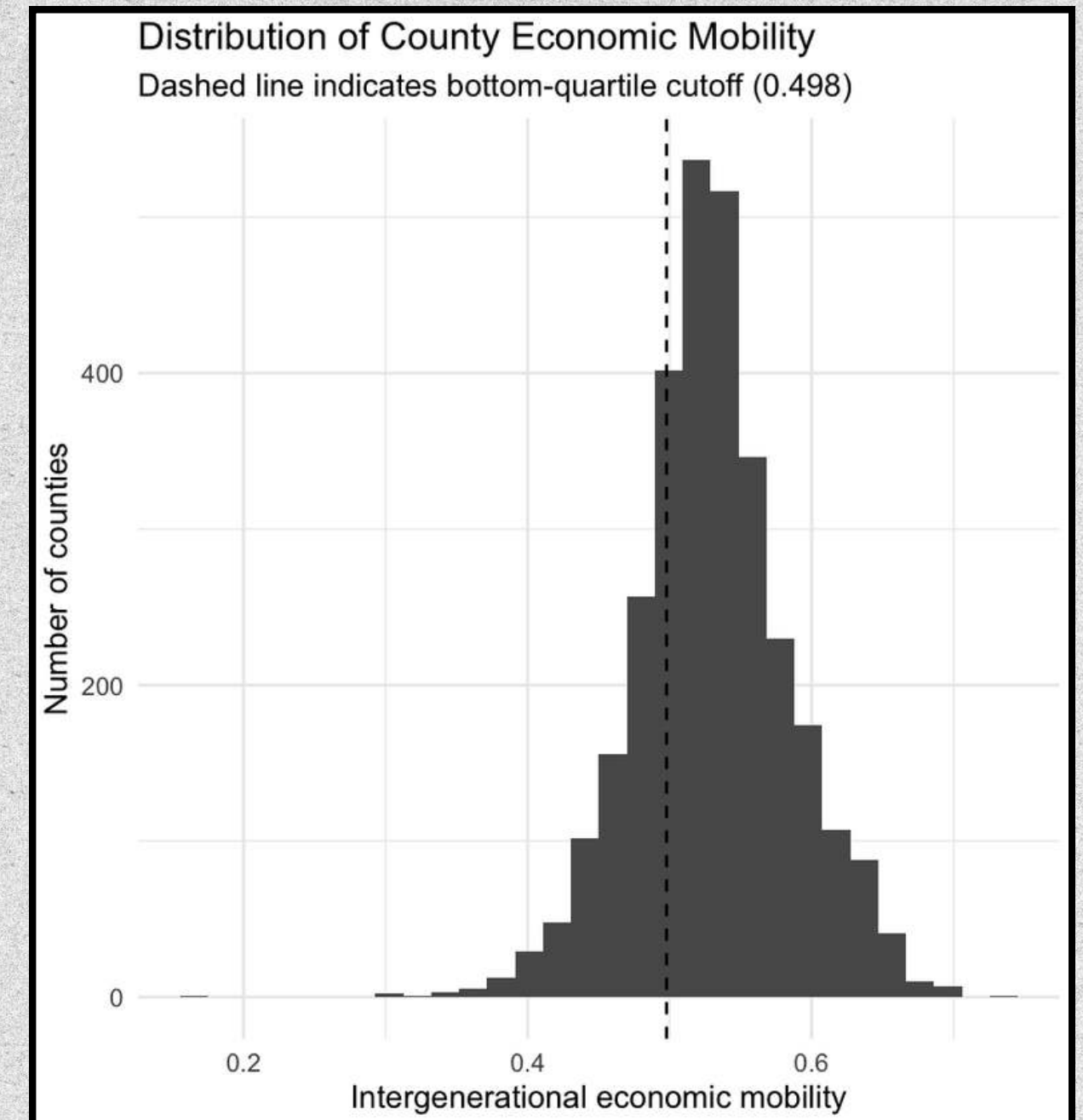
Where is economic opportunity most constrained?

How much can county-level social capital help explain and predict where economic mobility is high or low?



What exactly am I measuring?

- Primary outcome: continuous county economic mobility
- Secondary outcome: bottom 25% of national mobility distribution
- Low-mobility cutoff = 0.4978



Data Sources

The mobility outcome is historical: it averages cohorts born from 1978-1992.

That matters because today's county economy can look very different from the environment those children experienced growing up.

- 3 county-level datasets joined by FIPS
- 3,076 joined counties; 0 duplicate FIPS
- 3,011 counties in primary modeling sample
- 2,875 complete across all 15 cohorts

	County By Cohort Estimates
	County Names and FIPS
	Social Capital Data by County

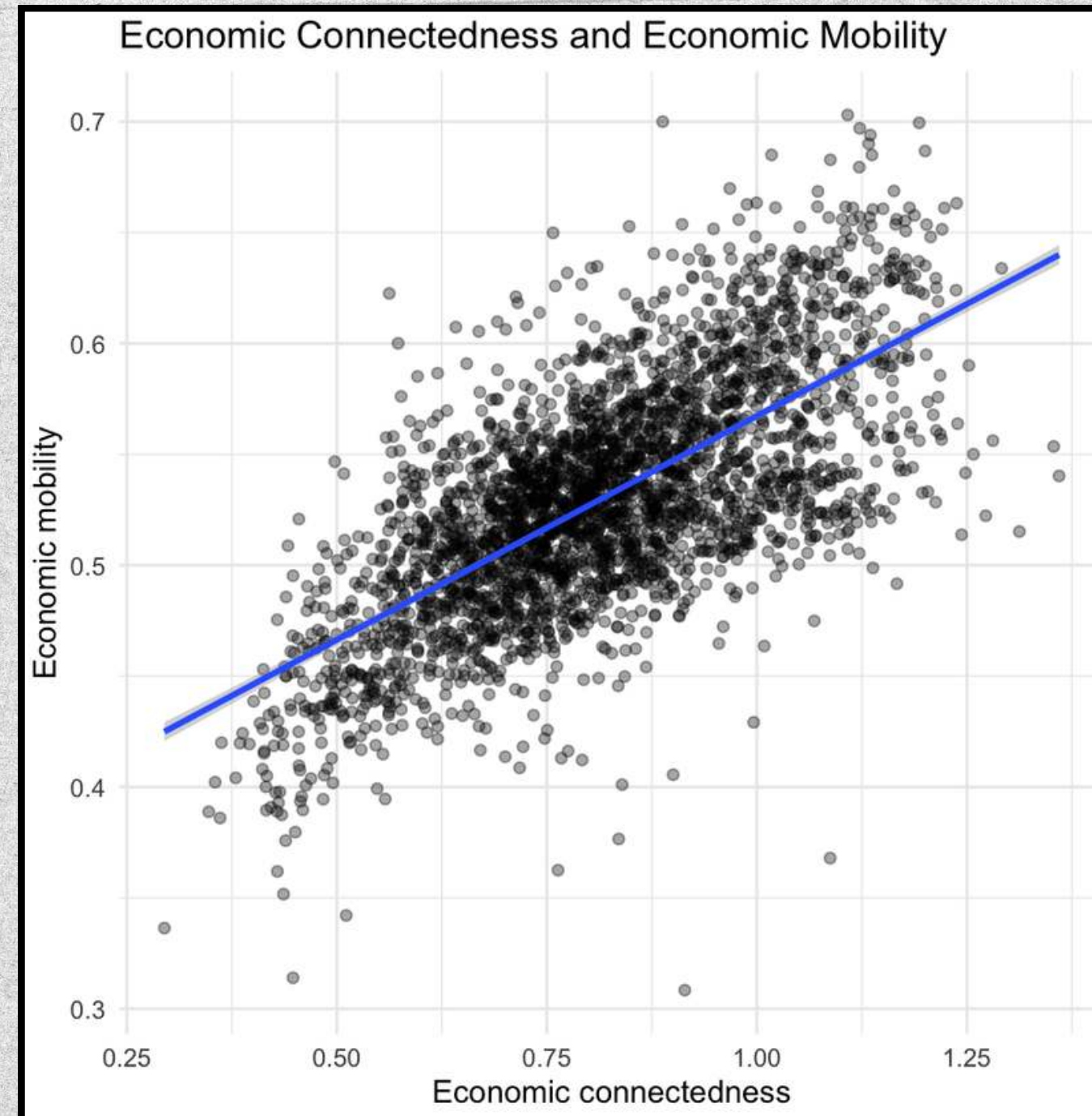
Economic Connectedness and Its Impact

This result lines up closely with Chetty et al. (Nature, 2022):

Economic connectedness (friendships that bridge socioeconomic groups) was one of the **strongest area-level predictors of upward mobility**, while several other social-capital measures were much less consistently associated.

This is reassuring external validation, not proof of causality.

- Economic connectedness vs mobility: $r = 0.68$
- Strongest bivariate social-capital relationship



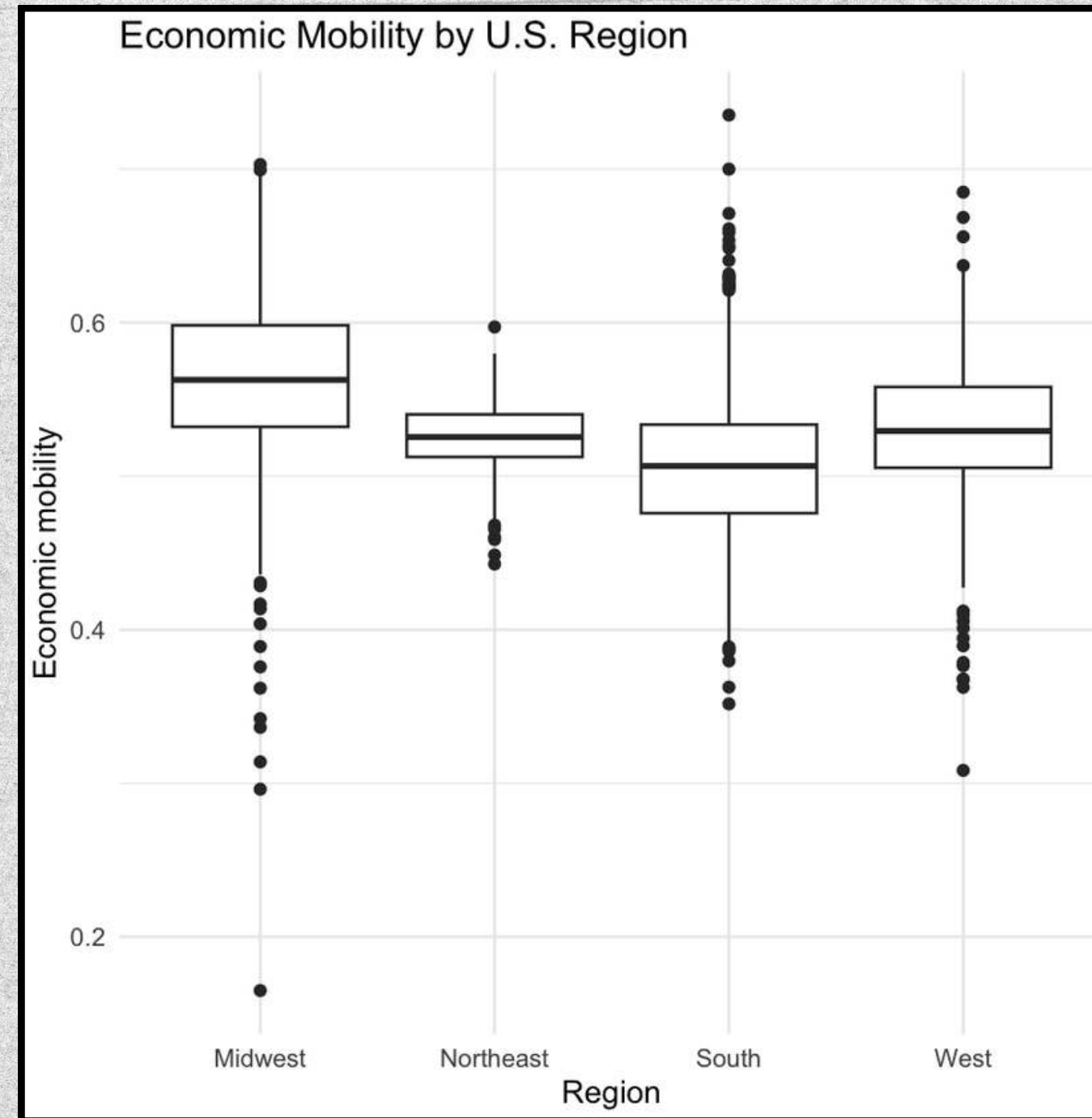
Other bivariate correlations with mobility:

- volunteering .33
- civic organizations .33
- support ratio .10
- clustering ~ 0
- log population $-.44$.

Geography Plays a Role

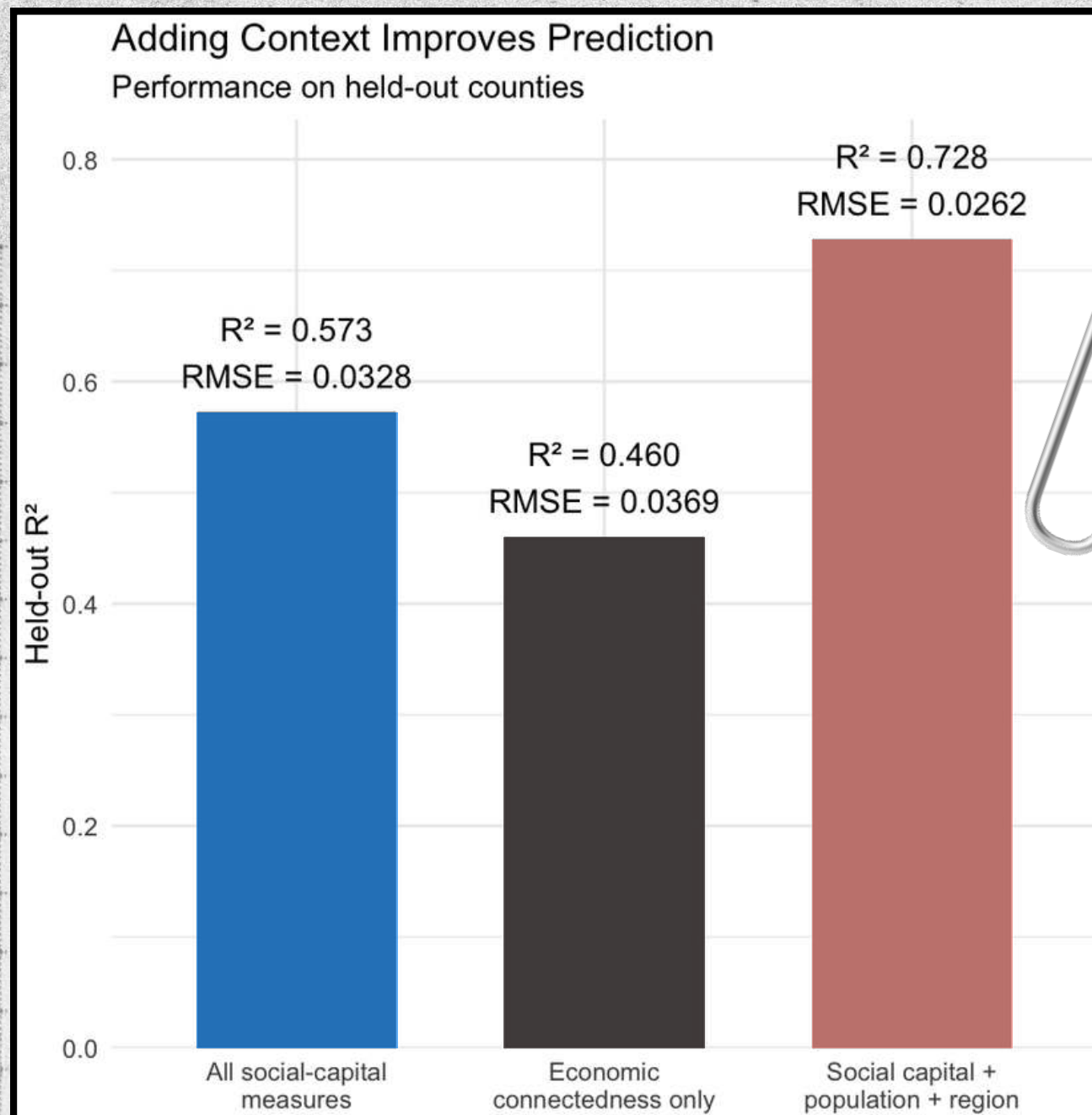
Prior mobility research finds that high-opportunity places tend to have less segregation and inequality, better schools, stronger social capital, and greater family stability.

- Mean mobility: Midwest .564; South .505
- Low-mobility share: South 42.7%
- Midwest 7.3% | Northeast 8.8% | West 18.7%



***Region is more of a broad proxy for structural differences the models presented later do not fully observe.**

model fitting



Model 1: EC only $\rightarrow$ test $R^2 = .460$

Model 2: all social capital $\rightarrow$ test $R^2 = .573$

Model 3: + population + region $\rightarrow$ test $R^2 = .728$

RMSE: .0369 $\rightarrow$.0328 $\rightarrow$.0262

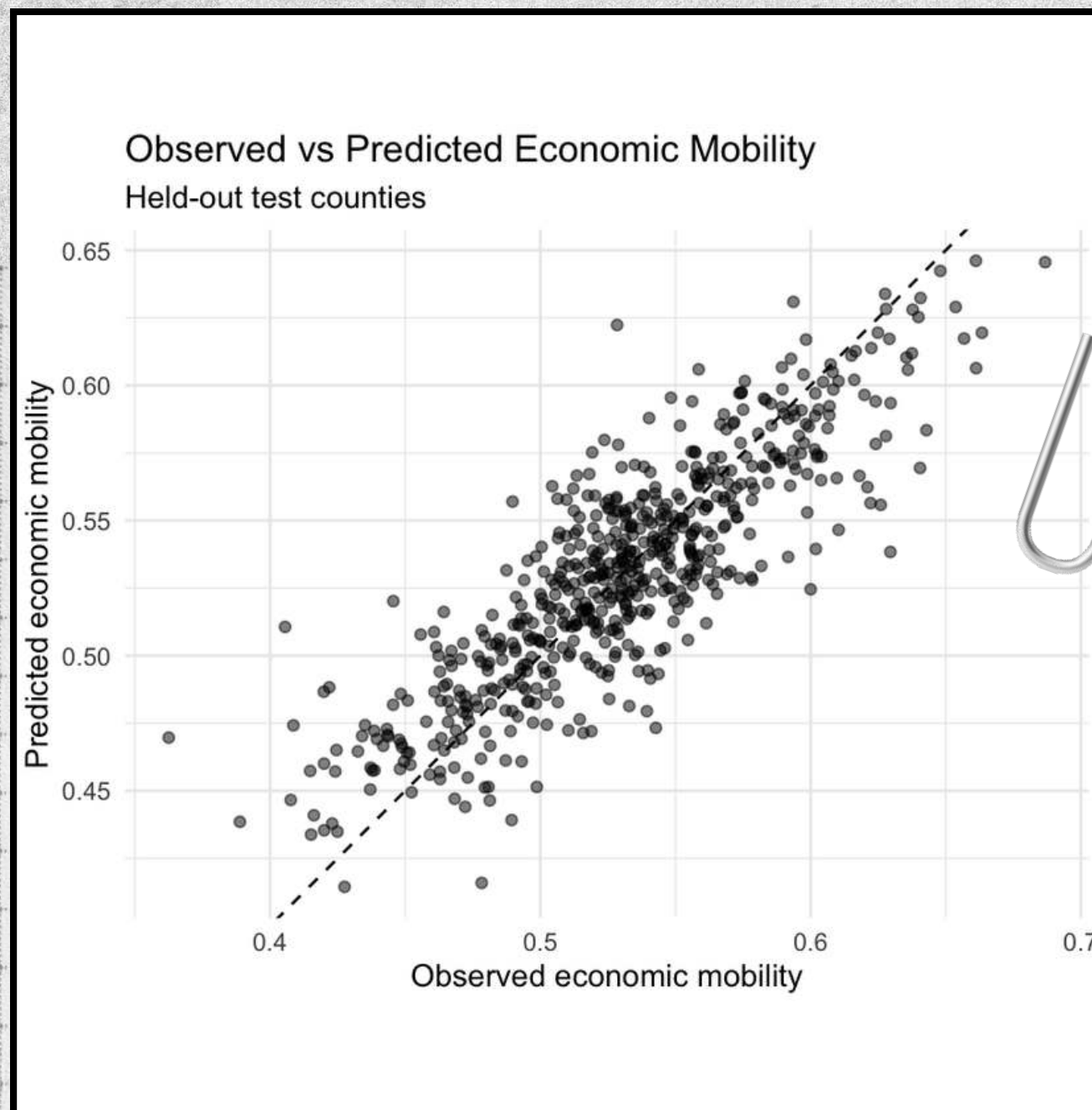
model fitting



**Midwest as the
reference group
for region:**

- 
- South vs Midwest: $\beta = -0.0201$, robust $p < .001$
 - West vs Midwest: $\beta = -0.0185$, robust $p < .001$
 - Northeast vs Midwest: $\beta = -0.0055$, robust $p = .011$
- 

model fitting



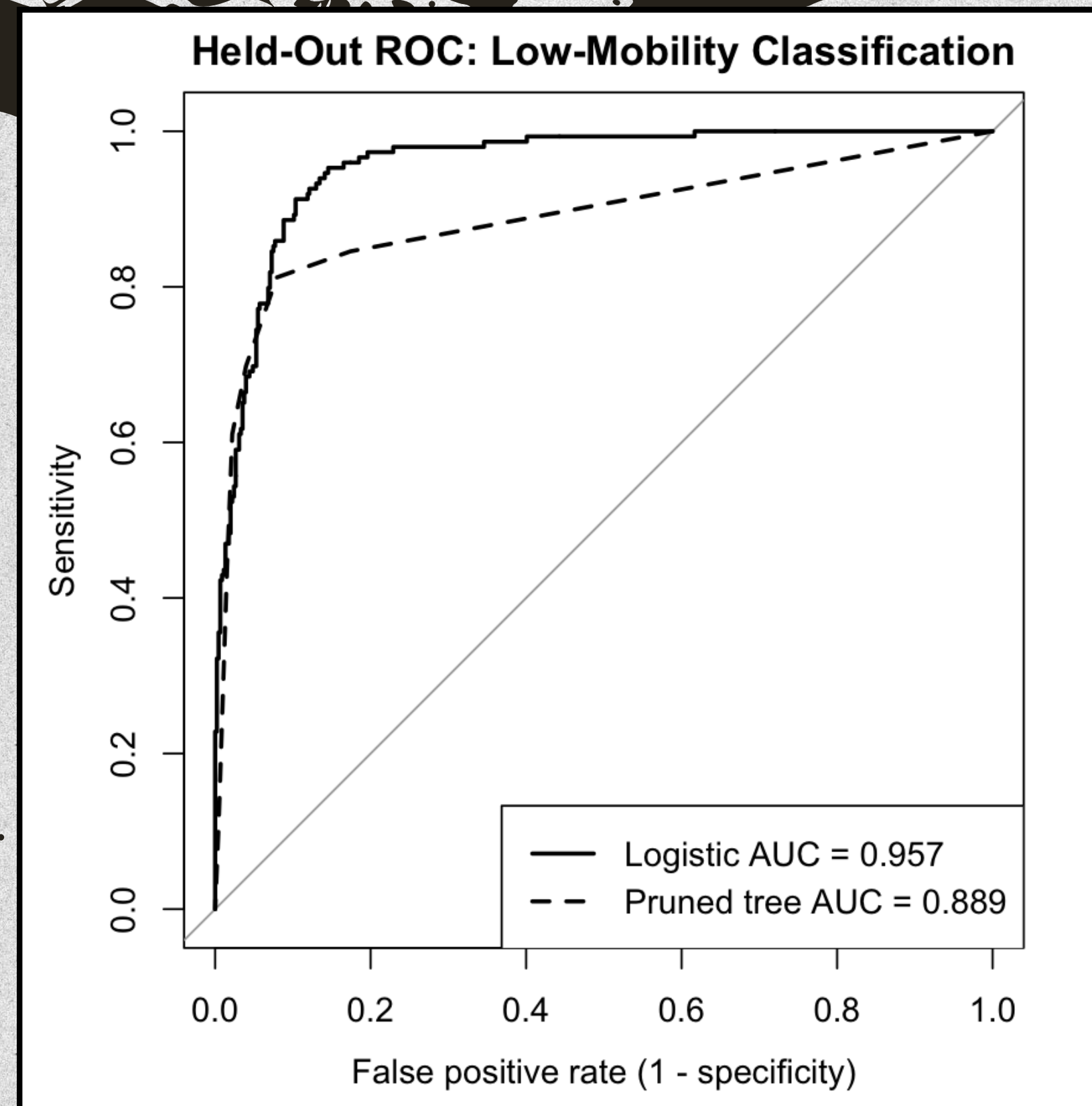
Predictions closely track observed mobility in held-out counties ($R^2 = .728$)

What does the full model say?

- Economic connectedness remained positively associated with mobility.
- Regional differences persisted after adjustment. South and West remain lower than Midwest
- Adding broader social-capital measures improved held-out R^2 from .460 to .573; adding population and region increased it further to .728.
- **Associations $\neq$ causal effects**

📍 Identifying the lowest-mobility counties

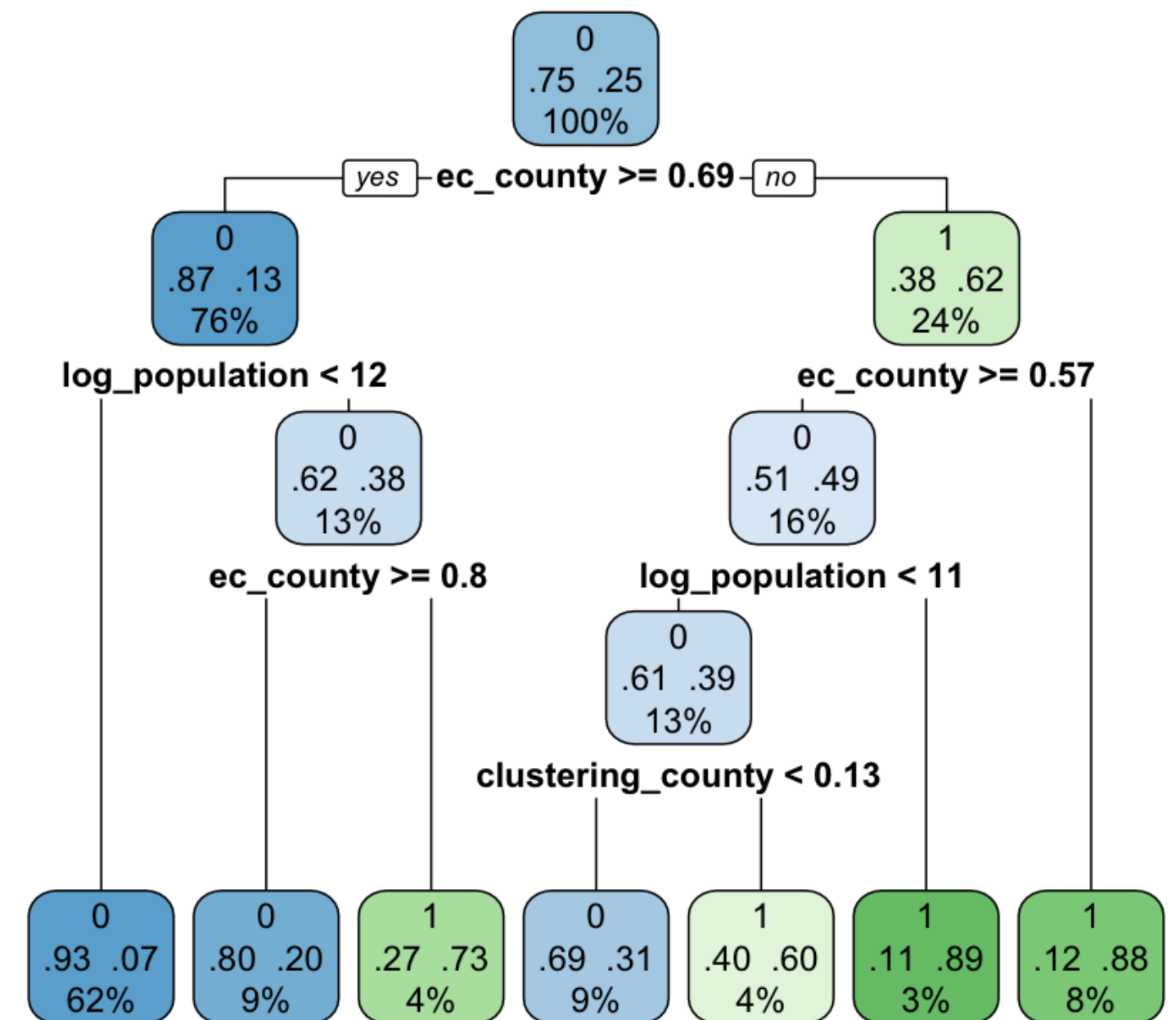
- Logistic regression: AUC .957 | Brier .073
- Pruned tree: AUC .889 | Brier .085
- Logistic model wins on ranking and probability accuracy



Decision tree makes pattern intuitive

- First split: economic connectedness ≈ 0.69
- Population and other features refine the split
- Tree is easier to read, logistic model predicts better

Pruned Decision Tree for Low-Mobility Counties



Conclusion

*Influential counties such as Storey, NV highlight the importance of local context.

Two caveats make the conclusion stronger: (1) today's economy is not the same thing as childhood mobility for 1978-1992 cohorts, and (2) county averages hide neighborhood variation.

Opportunity Atlas research shows large outcome differences even across nearby tracts within the same county.

- 15-cohort sensitivity: test $R^2 = .730^*$
- Primary model: test $R^2 = .728$
- Not causal; not a direct migration model
- **Takeaway: social capital + context strongly predict county opportunity**

Connectedness	Economic connectedness is strongly associated with higher mobility
Geography	Regional differences remain even after adjustment
Prediction	Social capital + population + region predict mobility well

**thank you for
listening!**

